

AI Based Thermal Powerline Hotspot Capstone Project

In this capstone, I will design an end-to-end AI pipeline to detect thermal hotspots in power lines and transmission towers using drone-based thermal inspection data.

I'm gonna use this dataset which has the following features -

`temp_mean`, `temp_max`, `temp_std`, `delta_to_neighbors`, `hotspot_fraction`, `edge_gradient`, `ambient_temp`, `load_factor`, `fault_label`

Next I am going to check for any missing data, upon further analysis, I see that there is no missing data in any of the columns.

Using Matplotlib and Seaborn to check for the number of abnormalities / `fault_label` in the data. And also generating a heatmap to understand the correlation between each feature. Warm ie red represents that both features rise together, and the Cold ie blue represents that when one feature rises the other one falls.



